

First Presentation – Optimization (Full Answers)

1. What is optimization? Give an example.

Optimization is the process of choosing decision variables in order to minimize or maximize a real-valued objective function, subject to given constraints on these variables. A classical example is maximizing profit

$$\max_{x_1, x_2 \geq 0} w_1 x_1 + w_2 x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq B,$$

where x_1, x_2 are production quantities, w_i are unit profits, and B is a budget.

2. Three main components of an optimization problem

A general single-objective optimization problem consists of:

1. **Decision variables** $x \in \mathbb{R}^n$ or in some discrete set.
2. **Objective function** $f(x)$ to be minimized or maximized.
3. **Constraints** $g_i(x) \leq 0$, $h_j(x) = 0$ (and possibly set constraints $x \in S$) describing feasible decisions.

3. Difference between decision variables, objective function, constraints

- **Decision variables** represent the unknown choices the decision maker can control, such as production quantities, flows, or design dimensions.
- The **objective function** maps each feasible choice x to a scalar value $f(x)$ measuring performance such as cost, profit, time, or energy, which we seek to minimize or maximize.
- **Constraints** are equations or inequalities that restrict which variable values are allowed, typically modeling resource limits, physical laws, or logical requirements.

4. Minimize versus maximize

To *minimize* $f(x)$ means to find a feasible x^* with

$$f(x^*) \leq f(x) \quad \forall x \text{ feasible},$$

while to *maximize* $f(x)$ means to find x^* with

$$f(x^*) \geq f(x) \quad \forall x \text{ feasible}.$$

Any maximization can be converted to minimization by minimizing $-f(x)$, and vice versa.

5. Feasible solution

A feasible solution is any assignment of the decision variables that satisfies all constraints of the optimization problem, including sign or integrality restrictions. In LP standard form, any x satisfying $Ax \leq b$ and $x \geq 0$ is feasible.

6. Infeasible solution

An infeasible solution is any assignment of the decision variables that violates at least one constraint, meaning it is not allowed by the problem's requirements. If no feasible solution exists, the whole problem is called infeasible.

7. What makes a solution optimal?

A feasible solution x^* is globally optimal if its objective value is best among all feasible points, that is, it attains the minimum or maximum over the feasible set. For minimization,

$$f(x^*) \leq f(x) \quad \text{for all feasible } x,$$

and for maximization the inequality is reversed.

8. Mathematical formulation of an optimization problem

A general nonlinear programming problem can be written as

$$\begin{aligned} \min_{x \in S} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i = 1, \dots, m, \\ & g_j(x) \leq 0, \quad j = 1, \dots, p, \end{aligned}$$

where $x \in \mathbb{R}^n$ is the decision vector, f is the objective function, h_i, g_j are constraint functions, and S is a given set. A linear program is the special case where f and all constraints are linear.

9. Linear vs. nonlinear objective functions

An objective $f(x)$ is *linear* if it has the form

$$f(x) = c^\top x = \sum_{j=1}^n c_j x_j,$$

that is, it is an affine function with no products or powers of variables beyond the first degree. It is *nonlinear* if it contains nonlinear terms such as products $x_i x_j$, powers x_i^2 , exponentials, and so on. Linear objectives lead to linear programming, while nonlinear objectives lead to nonlinear programming.

10. Feasible region

The feasible region, or feasible set, is the collection of all decision vectors that satisfy every constraint of the optimization problem. In a linear program, the feasible region is a convex polyhedron given by the intersection of halfspaces $Ax \leq b$ and possibly hyperplanes.

11. Solution space

The solution space, or decision space, is the full set of all possible values the decision variables can take before enforcing the constraints, typically a subset of $\mathbb{R}^n$ or a discrete set like $\{0, 1\}^n$. The feasible region is then a subset of this solution space determined by the constraints.

12. Solution space and feasible solution space of the given problem

Problem:

$$\begin{aligned} \max \quad & 5x_1 + 7x_2 + 2x_3 \\ \text{s.t.} \quad & 2x_1 + 4x_2 + x_3 \leq 4, \\ & x_1, x_2, x_3 \in \{0, 1\}. \end{aligned}$$

(a) **Solution space.** Because each variable is binary, the solution space is

$$\mathcal{X} = \{0, 1\}^3 = \{(x_1, x_2, x_3) \mid x_i \in \{0, 1\}, i = 1, 2, 3\},$$

which has $2^3 = 8$ possible points.

(b) **Feasible solution space.** We enumerate all 8 points and check the constraint $2x_1 + 4x_2 + x_3 \leq 4$:

(x_1, x_2, x_3)	$2x_1 + 4x_2 + x_3$	Feasible?
(0, 0, 0)	0	yes
(0, 0, 1)	1	yes
(0, 1, 0)	4	yes
(0, 1, 1)	5	no
(1, 0, 0)	2	yes
(1, 0, 1)	3	yes
(1, 1, 0)	6	no
(1, 1, 1)	7	no

Thus the feasible solution space is

$$\mathcal{X}_F = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (1, 0, 0), (1, 0, 1)\}.$$

If desired, one can also evaluate the objective on these feasible points to see that $(0, 1, 0)$ with value 7 is optimal, but the question only asks for the spaces.

13. Continuous vs. discrete optimization

In *continuous* optimization, the decision variables can take any real values in intervals or regions, for example $x \in \mathbb{R}^n$ with $x \geq 0$, and calculus-based methods and convex analysis are central. In *discrete* optimization, some or all variables are restricted to discrete sets such as integers or binaries, for example $x \in \{0, 1\}^n$, leading to combinatorial structures and requiring algorithms like branch-and-bound, cutting planes, or dynamic programming.

14. Combinatorial vs. calculus-based optimization

Calculus-based optimization assumes continuous variables and differentiable objective and constraint functions, and uses derivative information such as gradient, Hessian, KKT conditions, Newton, and quasi-Newton methods to search for optima. Combinatorial optimization, in contrast, deals with discrete structures such as graphs, sets, or integer vectors; the feasible set is finite or countable, derivatives are not defined, and solution methods rely on combinatorial algorithms such as branch-and-bound, dynamic programming, network flows, and heuristics.

15. Examples of real-life combinatorial optimization problems

Typical real-world combinatorial optimization problems include:

- **Traveling Salesman Problem (TSP):** find the shortest tour visiting each city exactly once and returning to the start.
- **Vehicle routing and logistics:** design routes for delivery trucks to minimize total distance or cost under capacity constraints.
- **Crew and shift scheduling:** assign staff to shifts or flights while respecting legal and contractual rules and minimizing cost.
- **Network design:** choose which links to build in a communication or power network to satisfy demand at minimum cost.
- **Set covering / facility location:** select facilities or sets to cover demand points with minimal total cost.

16. Linear programming formulations

(a) Multi-dimensional 0/1 knapsack problem

Data:

- n items, indexed by $j = 1, \dots, n$.
- Profit (value) p_j for item j .
- m resource constraints, capacities b_i for $i = 1, \dots, m$.
- Resource consumption a_{ij} of item j in dimension i .

Decision variables:

$$x_j = \begin{cases} 1, & \text{if item } j \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

Integer formulation:

$$\begin{aligned}
\max \quad & \sum_{j=1}^n p_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m, \\
& x_j \in \{0, 1\}, \quad j = 1, \dots, n.
\end{aligned}$$

If we relax integrality to $0 \leq x_j \leq 1$, we obtain the standard LP relaxation.

(b) Minimum set covering problem

Data:

- Universe $U = \{1, \dots, n\}$ of elements.
- Family of subsets $\mathcal{S} = \{S_1, \dots, S_m\}$ with cost c_j for set S_j .

Decision variables:

$$x_j = \begin{cases} 1, & \text{if set } S_j \text{ is chosen,} \\ 0, & \text{otherwise.} \end{cases}$$

Formulation:

$$\begin{aligned}
\min \quad & \sum_{j=1}^m c_j x_j \\
\text{s.t.} \quad & \sum_{j: i \in S_j} x_j \geq 1, \quad \forall i \in U, \\
& x_j \in \{0, 1\}, \quad j = 1, \dots, m.
\end{aligned}$$

Each element i must be covered by at least one selected set.

(c) Maximum independent set problem

Given an undirected graph $G = (V, E)$, an independent set is a subset of vertices with no edge between any two of them.

Data:

- Vertex set $V = \{1, \dots, n\}$.
- Edge set $E \subseteq \{\{i, j\} \mid i < j\}$.

Decision variables:

$$x_i = \begin{cases} 1, & \text{if vertex } i \text{ is in the independent set,} \\ 0, & \text{otherwise.} \end{cases}$$

Integer formulation:

$$\begin{aligned} \max \quad & \sum_{i \in V} x_i \\ \text{s.t.} \quad & x_i + x_j \leq 1, \quad \forall \{i, j\} \in E, \\ & x_i \in \{0, 1\}, \quad i \in V. \end{aligned}$$

The constraints ensure that no edge has both endpoints selected, so the chosen vertices form an independent set.